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Nonlinear dynamics of a corticothalamic coupled neural mass model reproduce the mechanism of 40 Hz ASSR generation

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ABSTRACT

Auditory steady-state responses (ASSR) provide a powerful tool for investigating oscillatory dynamics and network synchronization along the auditory pathway. However, existing neural mass models of 40 Hz ASSR have largely focused on cortical microcircuits or macroscopic phenomenology, while neglecting the closed-loop regulation formed by thalamocortical feedforward transmission and thalamic reticular nucleus (TRN)-mediated feedback inhibition. To address this, we constructed an Auditory Corticothalamic Neural Mass Model (ACT-NMM) that investigates cortical pyramidal neurons (Pyr), parvalbumin-positive interneurons (PV), somatostatin-positive interneurons (SST), thalamocortical relay neurons (TCR), and TRN neurons within a delay-coupled corticothalamic loop. By reproducing the steady-state oscillatory patterns observed in the auditory cortex and analyzing parameter-dependent synchronization, phase locking, and frequency locking, the ACT-NMM quantitatively characterizes how thalamocortical interactions shape ASSR dynamics. The results suggest that the auditory cortex acts as the primary generator of the 40 Hz ASSR, whereas the thalamus mainly contributes phase alignment and relay modulation, rather than generating a strong ASSR response independently. These findings demonstrate that ACT-NMM captures the key dynamic features of ASSR generation, providing a physiologically interpretable framework for linking corticothalamic parameters to observed auditory oscillations.

1. Introduction

Auditory steady-state response (ASSR) is defined as phase-locked neural oscillations that are induced by periodically amplitude or frequency-modulated auditory stimuli (Picton John and Dimitrijevic, 2003), reflecting the auditory system's ability to synchronize with continuous rhythmic input. Due to its robust frequency-following properties, the ASSR has become a widely used tool for probing oscillatory dynamics and network synchronization along the auditory pathway (Griskova Morup and Parnas, 2007) (King and Nelken, 2009). The 40 Hz ASSR engages a broad range of auditory pathways (Hackett et al., 2011) (Rubio et al., 2009). Recent findings (Zhao Tang and Chen, 2025) (Peng et al., 2021) (Li et al., 2020) (Roy et al., 2022) (Ibrahim Murphy and Yuditsev, 2021) suggest that thalamic inputs of distinct origins exhibit hierarchical, complementary, and highly specific

functional roles in shaping cortical auditory-evoked responses. Despite these advances, our understanding of the neurodynamic underlying 40 Hz ASSR remains limited.

Neural mass models (NMMs) characterize the collective behavior of neuronal populations by describing the dynamical evolution of their average firing rates (Breakspear, 2017) (Wilson and Cowan, 1973) (Liu et al., 2021). As ASSRs emerge from synchronized oscillatory activity across neural populations, NMMs and related population-level computational models are widely used to investigate auditory system dynamics, providing a tractable link between synaptic mechanisms and macroscopic electrophysiological responses. For example, Moran et al. (Moran Kiebel and Stephan, 2007) developed an NMM for spectral electrophysiological responses, establishing a frequency-domain description of steady-state membrane dynamics. Mill et al. (Mill Coath and Wennekers, 2011) reproduced stimulus-specific adaptation in the

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auditory cortex using interacting neuronal populations with inhibitory synapses. More recent studies have further examined how local excitation–inhibition (E–I) interactions and interneuron-specific dynamics shape auditory evoked responses within cortical circuit models. Collectively, these studies demonstrate the utility of population-level dynamical models for explaining various auditory phenomena. However, existing models have primarily focused on cortical local circuits, stimulus adaptation, or sensory gating mechanisms. The contribution of closed-loop thalamocortical regulation to the generation of 40 Hz ASSR, particularly the role of feedback inhibition mediated by the thalamic reticular nucleus (TRN), remains insufficiently explored.

The corticothalamic system is a nonlinear feedback network comprising interacting populations of excitatory and inhibitory neurons, whose dynamics are shaped by E–I balance, transmission delays, and synaptic activity. Experimental evidence suggests that 40 Hz ASSRs predominantly originate in the auditory cortex, while thalamic structures play a key role in shaping and sustaining synchronized oscillations. Reyes et al. (Reyes Lockwood and Salvi, 2005) localized the cortical generators of 40 Hz ASSRs using current density reconstruction. Park et al. (Park and Geffen, 2020) developed a cortical microcircuit model that incorporated PV and SST interneurons, demonstrating their distinct roles in auditory evoked responses; however, TRN and TRN-mediated feedback inhibition were not incorporated. Du et al. (D and Jansen, 2011) further investigated thalamocortical circuits and highlighted the importance of the TRN in sensory gating, yet their model did not distinguish between PV- and SST-specific inhibitory mechanisms, nor did it examine the parameter-dependent nonlinear dynamics underlying the 40 Hz ASSR. Therefore, a unified dynamical framework integrating thalamic relay transmission, TRN-mediated feedback inhibition, and PV/SST-dependent cortical E–I regulation is lacking. In particular, it remains unclear how these mechanisms jointly support stable, phase-locked 40 Hz oscillations, and which coupling strengths or synaptic time constants govern the emergence, maintenance, or suppression of ASSRs. To address these issues, we have proposed an auditory corticothalamic neural mass model (ACT-NMM) to investigate the dynamical mechanisms underlying the generation of 40 Hz ASSRs. The main contributions of this work are summarized as follows:

- (1) We formulate the auditory loop as a nonlinear dynamical system and reproduce the “thalamic relay-cortical amplification” mechanism of corticothalamic circuit. It effectively simulates the dynamical behavior of cortical ASSRs.
- (2) The corticothalamic coupling strength and time constant of cortical masses are taken as implicit control variables to analyze the ASSR-related bifurcation of ACT-NMM. The results show that cortical ASSRs emerge as a stable dynamic behavior only under periodic stimuli.
- (3) Through neural activity dynamical analysis of ACT-NMM, we demonstrate that ASSRs are mainly modulated in phase by the thalamus, and the response to 40Hz is the most significant under parameters that conform to physiological constraints.

By explicitly linking ASSR generation to dynamics of a delay-coupled nonlinear system, this study provides a system-level interpretation of 40Hz ASSR’s generation. Our results not only elucidate the dynamical mechanisms underlying ASSRs, but also offer a principled computational framework to link observed neural dynamics with underlying parameters of neural mass micro-circuit.

2. Methods

The modeling is based on the hypothesis that intrinsic 40 Hz oscillations within the auditory neural network synchronize with 40 Hz components in auditory stimuli. The ACT-NMM reconstructs the intrinsic dynamics of cortical ASSRs from neurophysiological variables, such as membrane potentials, synaptic strengths, and time constants.

2.1. Thalamic node: TCR-TRN two-population model

In the auditory circuit, auditory information flows from the thalamus (exactly MGB) to the auditory cortex, and the MGB nucleus receives inhibitory feedback sent by the TRN to the thalamic relay neurons (TCR). Early studies indicated that the thalamus contains only inhibitory interneurons (Jones, 2007). Therefore, the thalamic rhythm relies on two functionally complementary neural populations: TCR and the TRN. Besides, Li et al. (Li Li and Wang, 2024) demonstrated in awake mice that the generation of cortical ASSR critically depends on the corticothalamic projections from GABAergic interneurons in the ventral division of the medial geniculate body to the granular layer of the auditory cortex, and that this process is tightly regulated by TRN activity. These findings provide a theoretical basis for modeling thalamic contributions to the corticothalamic projections during 40 Hz ASSR using TCR and TRN populations. Accordingly, this study constructs a thalamic node of the auditory circuit based on the TCR-TRN two-population thalamic NMM purposed by Costa et. al. (Schellenberger Costa Weigenand and Ngo, 2016). The dynamic equations of the thalamic node are as follows.

$$I_h = g_h(m_{h1} + g_{inc}m_{h2})(V_{TCR} - E_h) \quad (1)$$

$$I_L = g_L(V - E_L) \quad (2)$$

$$I_{LK} = g_{LK}(V - E_K) \quad (3)$$

$$I_T = g_T(m_{inf,T})^2 h_T(V_K - E_{Ca}) \quad (4)$$

$$s_{e/i}(t) = \sum_k \alpha_{e/i}(t) \otimes N_{kk} r_k(t) + \Phi(t) \quad (5)$$

$$I_{AMPA} = g_{AMPA} s_e(V - E_{AMPA}) \quad (6)$$

$$I_{GABA} = g_{GABA} s_i(V - E_{GABA}) \quad (7)$$

$$\frac{dV_{TCR}}{dt} = \frac{I_L^{TCR} + w_{ee}I_{AMPA}^{TCR} + w_{ei}I_{GABA}^{TCR}}{\tau_{TCR}} - \frac{I_{LK}^{TCR} + I_T^{TCR} + I_h}{\tau_{TCR}C_m} \quad (8)$$

$$\frac{dV_{TRN}}{dt} = \frac{I_L^{TRN} + w_{ie}I_{AMPA}^{TRN} + w_{ii}I_{GABA}^{TRN}}{\tau_{TRN}} - \frac{I_{LK}^{TRN} + I_T^{TRN}}{\tau_{TRN}C_m} \quad (9)$$

$$fr_{TCR/TRN} = \frac{f_{max}}{1 + e^{\frac{C_1 \cdot (V_{TCR/TRN} - \theta_{th})}{\sigma}}} \quad (10)$$

$$\frac{dfr_{TCR/TRN}}{dt} = -\lambda(fr_{TCR/TRN}(t) - fr_{TCR/TRN}(t-1)) \quad (11)$$

where g_h , g_L , g_{LK} , and g_T represent the hyperpolarization-activated cation non-selective conductance, the leakage conductance, the potassium leakage conductance, and the T-type calcium conductance respectively; and $g_{AMPA/GABA}$ represents the excitatory/inhibitory receptor conductance. E_h , E_L , E_K , and E_{Ca} indicate the Nernst reversal potential of the hyperpolarization-activated cation, the leakage Nernst reversal potential, the Nernst reversal potential of potassium, and the Nernst reversal potential of calcium respectively, while $E_{AMPA/GABA}$ indicates the Nernst reversal potential of the excitatory/inhibitory receptor. I_h is the hyperpolarization-activated cation non-selective current. m_{h1} , m_{h2} are the gating functions. g_{inc} is the conductance scaling factor. I_L is the leakage current, and I_T is the T-type calcium current. $m_{inf,T}$ and h_T are the gating functions of the T-type calcium current. $s_{e/i}$ represents the synaptic efficiency of AMPA/GABA, $\alpha_{e/i}(t) = \gamma_{e/i}^2 \exp(-\gamma_{e/i}t)$ is the synaptic response to a single spike. The average

connection strength between TCR and TRN is $W = \begin{bmatrix} w_{ee} & w_{ei} \\ w_{ie} & w_{ii} \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ 3 & 5 \end{bmatrix}$. N_{kk} represents the connection strength between presynaptic k

and postsynaptic k' , r_k is the spike rate of k' , Φ is background noise, λ is the firing rate convergence rate constant of neurolib.

The ion current mechanism of the thalamic node model given by Eqs. (1)-(4) realizes the rebound bursting of spindle oscillations. g_L and g_{LK} are equally on TCR and TRN, while g_T differ between neuron populations. Eqs. (8)-(9) realizes the update of the average membrane potential of neurons. The Eq. (10) realizes the calculation of the firing rate, and the average firing rate of TCR is used as the input signal transmitted from the thalamus to the cortex to drive cortical neuron populations.

2.2. Cortical node: pyr-PV-SST three-population model

The mammalian sensory cortex comprises diverse excitatory and inhibitory neurons that form complex microcircuits for processing sensory information. Prior modeling work has emphasized the functional roles of these interneuron subtypes. For example, Aponte et al. (Aponte Handy and Kline, 2021) developed a three-population model incorporating Pyr, PV, and SST, demonstrating that modifying SST connectivity or excitatory drive can reproduce inhibitory imbalance and direction-selective modulation in A1. Phillips et al. (Phillips Schreiner and Hasenstaub, 2017) further showed that PV and SST interneurons play a key role in shaping auditory-evoked responses, temporal adaptation, and network dynamics. Experimental recordings in anesthetized mice also revealed that cortical activity closely tracks thalamic input and linearly amplifies them (Li et al., 2013) (Lien and Scanziani, 2013).

We have considered the Pyr, PV, and SST populations in the cortical node, where Pyr and PV receive thalamic input, while SST is driven solely by Pyr activity. PV and SST realize fast/slow inhibitory regulation of Pyr, and the mechanism of synaptic resource attenuation is considered during the process of receiving thalamic input. The standardized firing rate is adopted as corticothalamic coupling variables, the dynamic equations of the cortical node are as follows.

$$\begin{bmatrix} I_{Pyr} \\ I_{PV} \\ I_{SST} \end{bmatrix} = \begin{bmatrix} w_{ee} & w_{ep} & w_{es} \\ w_{pe} & w_{pp} & w_{ps} \\ w_{se} & w_{sp} & w_{ss} \end{bmatrix} \begin{bmatrix} fr_{Pyr} \\ -fr_{PV} \\ -fr_{SST} \end{bmatrix} + \begin{bmatrix} q \\ q \\ 0 \end{bmatrix} gfr_{TCR} \quad (12)$$

$$\tau_{Pyr} \frac{dr_{Pyr}(t)}{dt} = -r_{Pyr}(t) + I_{Pyr}(t) \quad (13)$$

$$\tau_{PV} \frac{dr_{PV}(t)}{dt} = -r_{PV}(t) + I_{PV}(t) \quad (14)$$

$$\tau_{SST} \frac{dr_{SST}(t)}{dt} = -r_{SST}(t) + I_{SST}(t) \quad (15)$$

$$\frac{dg(t)}{dt} = \frac{g_0 - g(t)}{\tau_{d1}} - \frac{g(t)fr_{TCR}(t)}{\tau_{d2}} \quad (16)$$

$$fr_a = \frac{f_{max}}{1 + e^{-r(r_a - a_{th})}} \quad (17)$$

$$\frac{dfr_a(t)}{dt} = -\lambda(fr_a(t) - fr_a(t-1)) \quad (18)$$

where I_a is the total postsynaptic input firing rate ($a = Pyr, PV, SST$), q represents the strength of thalamocortical input, and g is the dynamic transmission gate of thalamic input. fr_a is the standardized firing rate, τ_a is the membrane time constant, r_a is the activity-like potential based on corticothalamic coupling, $f(r_a)$ represents the firing rate function, f_{max} is the maximum firing rate, r is the slope of firing rate function, a_{th} is the activity threshold, and λ is the firing rate convergence constant of neurolib.

This study adopts the connection weight matrix W , considering that the average connection strength of neuron populations remains fixed throughout the induction of auditory responses (Mill et al., 2011). Based on the conclusions of physiological experiments and the parameter settings of previous studies (Li et al., 2014) (Pfeffer et al., 2013), the

average connection strength between neuron populations is determined as follow.

$$W = \begin{bmatrix} w_{ee} & w_{ep} & w_{es} \\ w_{pe} & w_{pp} & w_{ps} \\ w_{se} & w_{sp} & w_{ss} \end{bmatrix} = \begin{bmatrix} 1.1 & 2.0 & 1.0 \\ 1.0 & 2.0 & 2.0 \\ 6.0 & 0.0 & 0.0 \end{bmatrix} \quad (19)$$

According to the current compensation mechanism proposed by Park et al. (Park and Geffen, 2020), PV actively compensate for the missing inhibition of SST, while SST will passively enhance in the late stage of stimulation to assist in maintaining a steady-state E-I balance. In Eq. (16), the dynamic thalamic input transmission gate is used to model the attenuation of synaptic resource during the cortex receiving thalamic input. Physiologically, the dynamic transmission gate $g(t)$ is introduced to represent the short-term reduction in the effectiveness of thalamocortical drive during sustained rhythmic input. Continuous TCR firing consumes available synaptic resources, gradually reducing the effective feedforward impact on the cortical Pyr and PV populations. Reduced input, however, allows partial recovery of the gate. Therefore, $g(t)$ does not replace the anatomical thalamocortical connection weight, but rather provides a state-dependent gain term that prevents unrealistically persistent cortical excitation and mimics the transient-to-steady-state adaptation commonly observed in sensory-evoked thalamocortical responses. This modeling choice is also consistent with the functional roles of PV and SST inhibition: PV provides rapid early suppression of Pyr activity, while SST contributes slower inhibitory stabilization during sustained stimulation. Together, they maintain the E-I balance required for stable ASSR generation.

2.3. ACT-NMM framework

This study uses the aforementioned thalamic node and cortical nodes to build an ACT-NMM. Fig. 1 shows the model structure of the corticothalamic coupling framework. It consists of two nodes: the cortex and the thalamus, which form an overall network. By combining the aforementioned delay dynamic equations with the forward Euler method, the fast generation of neural mass firing activities is enabled through just-in-time (JIT) compilation. The simulation integration time step $dt = 0.1$ ms. The numerical modeling of corticothalamic coupling in the ASSR task state based on 40 Hz chirp signal input is realized. The cortical node has both feedforward and feedback connections; however, only the TRN of the thalamic node has self-feedback connections because thalamic relay cells typically do not form local connections within the population (Jones, 2001). There are bidirectional, long-range afferent connections between the Pyr of cortex and the TCR of thalamus. The delay of these long-range projections is set to 13 ms (D and Jansen, 2011) (Roux et al., 2013).

As a core parameter of membrane potential dynamics, the neuronal time constant directly determines how quickly neurons respond to input signals and their ability to synchronize oscillations. To avoid the model deviating from physiological significance due to the arbitrariness of the parameter range, its value must strictly follow the results of biophysical experimental observations. When combined with neural physiological experimental data and theoretical analysis, the time constants of each neuron population must satisfy the physiological constraints listed in Table 1, ensuring that the model can accurately reproduce the neural mechanism of ASSR.

The coupling strength determines the efficiency of information transmission. If the coupling is excessively weak, signal transmission between the cortex and thalamus will be hindered, making it impossible to integrate sensory information (e.g. the hierarchical processing of visual and auditory signals). In ACT-NMM, interaction between the cortex and thalamus occurs through coupling of neuron populations. Bidirectional regulatory $w_{thal-cx}$ and $w_{cx-thal}$ effects are introduced into fr_{TCR} and fr_{Pyr} via additive coupling. The grid search scanning range of coupling strength is set to 0~5 (step size is 0.5).

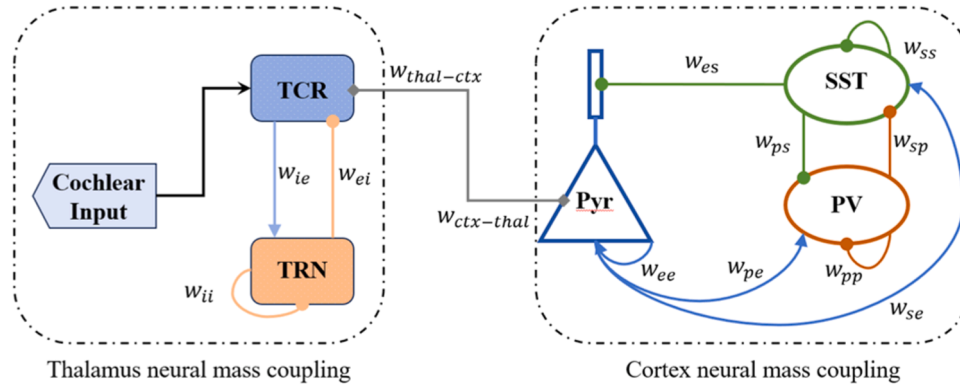


Fig. 1. Schematic diagram of ACT-NMM.

Table 1

Physiological constraints.

Neurons	Constraint Basis	Range
Pyr	L5 of rat sensory cortex: 16 ± 5.3 ms (Koch, Rapp and Segev, 1996) (Cauller and Connors, 1994)	10-20 ms
PV	Rat sensory cortex: 7.7 ± 3.8 ms (Povysheva, Zaitsev and Rotaru, 2008)	5-15 ms
SST	Rat prefrontal cortex: 20.5 ± 6.5 ms (Halabisky et al., 2006)	15-25 ms
TCR	Rat: 17 ± 1 ms (Gabernet, Jadhav and Feldman, 2005) Ferrets: 11.8 ± 0.8 ms (Alexander, Fisher and Godwin, 2006)	10-20 ms
TRN	Ferrets: 4.9 ± 0.5 ms (Alexander, Fisher and Godwin, 2006)	5-10 ms

2.4. Stimulus conditions and analysis methods

Two types of stimuli (40 Hz chirp and Ornstein-Uhlenbeck (OU) process noise) were set up to verify the model's specific response to 40 Hz periodic stimuli. To ensure a rigorous comparison between the stochastic background activity and the deterministic 40 Hz Chirp (amplitude is 0.2), we set offset $\mu = 0$ and decay time constant $\tau = 25$ ms to reflect physiological synaptic integration times. The standard deviation $\sigma = 0.03$ ensuring that 95% of the noise fluctuations remain within the same range as the Chirp stimulation. The evaluation indicators used in this study were defined as follows. The phase-locking value (PLV) quantifies the consistency of the instantaneous phase difference between two signals. It is calculated as follows:

$$PLV = \frac{1}{N} \left| \sum_{n=1}^N \exp[j(\varphi_x(n) - \varphi_y(n))] \right| \quad (20)$$

where $\varphi_x(n)$ and $\varphi_y(n)$ are the Hilbert phases of the two signals. PLV ranges from 0 to 1; larger values indicate stronger phase synchronization.

The coefficient of variation (CV) measures the relative fluctuation of oscillatory amplitude and is calculated as follows:

$$CV = \frac{\sigma_A}{\mu_A} \quad (21)$$

where σ_A and μ_A are the mean and standard deviation of the amplitude envelope A , respectively. A smaller CV indicates a more stable steady-state response.

The frequency deviation quantifies frequency-locking accuracy and is calculated as follows:

$$\Delta f = |f_{peak} - f_{stim}| \quad (22)$$

where f_{peak} is the dominant spectral peak of the simulated neural activity and f_{stim} is the stimulation frequency. Smaller Δf therefore indicates stronger frequency locking to the auditory stimulus.

In terms of evaluation metrics, the frequency-locking ability is verified through the frequency deviation (values below 2 Hz) and the target frequency power ratio (values above 0.3). The synchronization involves calculating the PLV of $f_{TCR} - f_{Pyr}$. The CV and time-frequency power is verified the time-frequency stability of 40 Hz response.

3. Results

As existing studies have not quantified the connection between electrophysiological signals and neuron firing activity, this paper uses the f_{Pyr} as source signal for ASSR state analysis. The state mapping relationship between the neuron firing rate and ASSR amplitude and power is not considered.

3.1. Parameter sensitivity analysis

To evaluate the effect of ACT-NMM in reproducing the related neural activity patterns, the boundary conditions of 40 Hz ASSR are analyzed from two aspects: coupling strength and the time constant of cortical populations, respectively. The neuron parameters of the model are shown in Table 2, which are based on related research (Park and Geffen, 2020) (D and Jansen, 2011) (Kim Jeong and Lee, 2016) (Malik Pai and Rubin, 2019). A comprehensive analysis was performed to determine the impact of corticothalamic coupling strengths on the 40 Hz ASSR. We repeated 20 times numerical simulation for each combination and took the average to reduce the randomness of results. As illustrated in Fig. 2, we conducted a bifurcation analysis on four indicators, namely thalamic ASSR activation, cortical ASSR activation, thalamocortical 40 Hz PLV, and global E-I firing rate balance, based on the bidirectional corticothalamic coupling strength $w_{thal-ctx}$ and $w_{ctx-thal}$, and visualized the

Table 2

ACT-NMM parameters.

Parameter	Value	Parameter	Value
τ_{Pyr}	15.0 ms	τ_{TCR}	12.0 ms
τ_{PV}	5.0 ms	τ_{TRN}	6.0 ms
τ_{SST}	21.0 ms	θ_{th}	-58.5 mV
r	3.0	σ	6.0
Pyr_{th}	0.7	C_1	1.8137993642
PV_{th}	1.0	C_m	1.0
SST_{th}	1.0	g_L	1.0 ms/cm ²
f_{max}	150 Hz	g_{LK}	0.051 ms/cm ²
τ_{d1}	1500.0 ms	g_{TCR}^{TCR}	3.0
τ_{d2}	20.0 ms	g_{TRN}^{TRN}	2.3
g_0	1.0 ms/cm ²	g_h	0.040 ms/cm ²
q	5.0	E_{AMPA}	0 mV
g_{AMPA}	1.0 ms/cm ²	E_{GABA}	-70 mV
g_{GABA}	1.0 ms/cm ²	E_L	-70 mV
$w_{thal-ctx}$	3.0	E_K	-100 mV
$w_{ctx-thal}$	2.0	E_{Ca}	120 mV

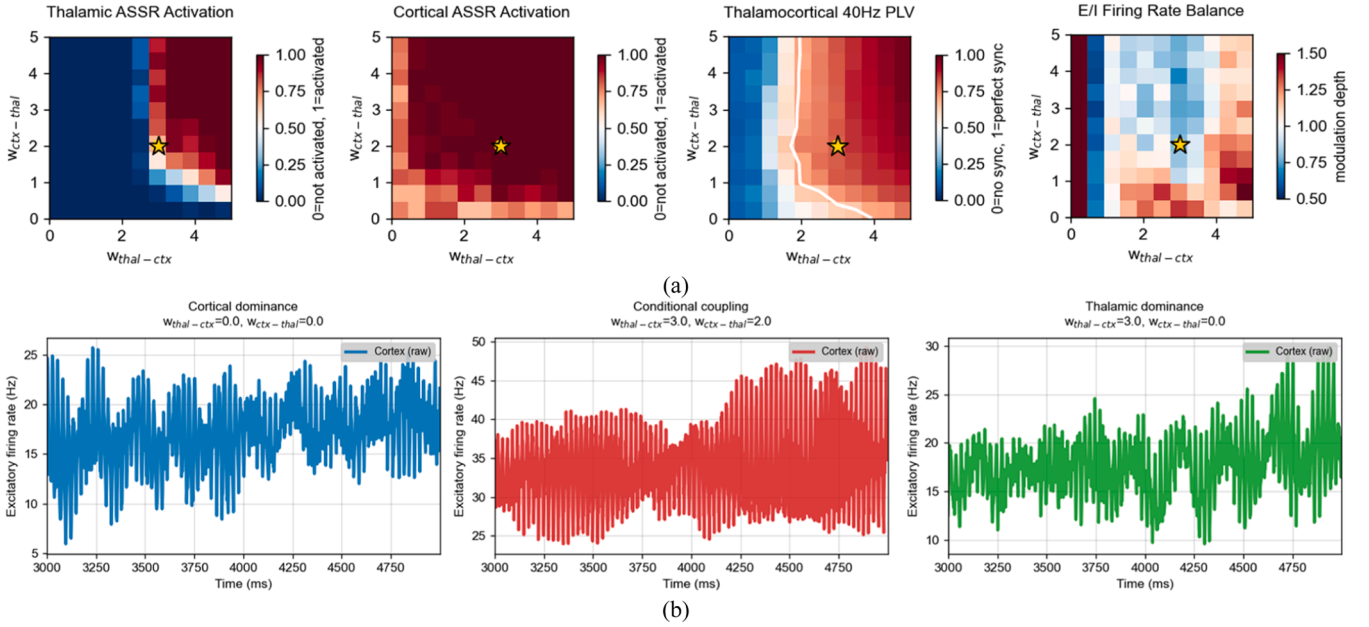


Fig. 2. Bifurcation dynamic of corticothalamic coupling strength. (a) Thalamic ASSR activation, cortical ASSR activation, thalamocortical 40 Hz PLV, and excitation–inhibition (E/I) firing-rate ratio across different corticothalamic coupling strengths. According to the PLV bifurcation analysis, coupling states can be categorized into three dynamical regimes: low-locking state ($PLV < 0.3$), boundary-locking state ($0.3 \leq PLV < 0.7$), and high-locking state ($PLV \geq 0.7$). The emergence of the white boundary ($w_{thal-ctx} \geq 2.0$ & $w_{ctx-thal} \geq 1.0$) therefore marks the transition from weak or intermediate synchronization to robust thalamocortical phase locking. (b) The oscillation of fr_{pyr} under three typical values of corticothalamic coupling strength.

cortical Pyr firing patterns of three typical values.

It can be found in Fig. 2 (a) that the thalamic ASSR activates after the boundary ($w_{thal-ctx} = 3.0, w_{ctx-thal} = 2.0$), while the cortical ASSR is widely activated. Besides, the thalamocortical 40 Hz PLV shows a high phase-locking boundary when $w_{thal-ctx} \geq 2.0$. Meanwhile, a severe imbalance in the E-I firing rate occurs when $w_{thal-ctx} \leq 1.0$, which is significantly improved as its increases. This indicates that the corticothalamic E-I balance of ACT-NMM has a fundamental strength requirement for $w_{thal-ctx}$. Fig. 2 (b) shows three typical firing patterns of cortical Pyr that appear in the bifurcation of coupling strength. First, under the condition of no coupling, the Pyr discharge is completely dominated by the intrinsic dynamics of the cortex. The subscripts baseline, fr_{pyr} oscillate between 5–25 Hz with 15 Hz as the baseline. Based on the thalamic activation boundary identified in Fig. 2 (a), we also visualized the fr_{pyr} oscillations of the boundary point ($w_{thal-ctx} = 3.0, w_{ctx-thal} = 2.0$) whose baseline level is between 30–40 Hz. When $w_{thal-ctx} \geq 2.5$, the thalamus begins to modulate the firing characteristics of cortical Pyr. The baseline level of fr_{pyr} oscillations is closer to the 40 Hz stimulation frequency and enters a steady-state firing mode more quickly. After a comprehensive analysis of the parameter sensitivity of corticothalamic coupling strengths, we believe that the boundary point can effectively activate the 40 Hz ASSR. It will continue to be used in subsequent analyses to avoid high coupling levels from interfering with the results of sensitivity analyses of other parameters. Although the selected time constant combination for 40 Hz ASSR can be determined from simplified evaluation indicators, activity records under a single task/input may still exhibit multiple parameter combinations that correspond to the same activity (Beiran and Litwin-Kumar, 2025). The impact on the neurodynamic must be determined further through biological constraints combined with the parameter dynamics analysis.

To unravel the dynamic mechanisms underlying these coupling effects, we categorized the system's behavior into three distinct regimes based on the thalamocortical 40 Hz PLV. Specifically, we divided different corticothalamic coupling strengths combinations ($w_{thal-ctx}, w_{ctx-thal}$) into three regimes: high locking state ($PLV \geq 0.7$), boundary locking state ($0.3 \leq PLV \leq 0.7$), and low locking state ($PLV \leq 0.3$). As

shown in Fig. 3, the average amplitude stability and E/I ratio are the best in the high phase-locking state, and the average $PLV = 0.851$ indicates that thalamic nodes effectively modulate the phase of Pyr. By comparing high locking state and boundary locking state, the cortical Pyr can still enter a steady-state periodic discharge process after the strong phase modulation of thalamus effectively. The thalamus effectively modulates the phase of Pyr through its relay function, enabling the cortex to achieve stable periodic firing. In the low locking state, although the thalamus does not effectively modulate the phase, the cortex still exhibits periodic oscillations (with a small CV), indicating that the Pyr-PV-SST three-population microcircuit within the cortical nodes itself has intrinsic dynamic characteristics to maintain basic oscillations, which confirms the conclusion that "the cortex is the main generator of 40Hz ASSR". The E-I ratio is positively correlated with PLV, and in the high-locking state, the E-I ratio = 0.992 is close to the ideal balance value, suggesting that thalamocortical coupling maintains the cortical E-I balance by regulating the activity of inhibitory neurons (PV and SST), which provides a basis for stable oscillation of fr_{pyr} .

To comprehensively demonstrate the impact of the time constant combinations of the three cortical neuron populations on the cortical 40Hz ASSR, we performed a traversal sampling of parameter combinations at 1 ms intervals within the value ranges of τ_{pyr} , τ_{pv} , and τ_{sst} determined in Table 1. For each parameter combination, a 5000 ms numerical simulation was conducted uniformly to evaluate the phase-locking ability and amplitude variability of fr_{pyr} under different ($11 \times 11 \times 11 = 1331$) cortical time constant conditions. The CV of fr_{pyr} amplitude and thalamocortical PLV for different cortical time constant combinations were visualized in Fig. 4. We also visualized in the figure the convex hulls corresponding to the top 30 combinations with "high PLV and low CV", providing a clear range of cortical time constants for subsequent analyses. When $\tau_{pyr} < 13ms$, the PLV decreases significantly (phase synchronization ability weakens); when $\tau_{pyr} > 16ms$, CV increases (amplitude stability deteriorates). This indicates that the membrane time constant of pyramidal neurons directly determines the response speed and oscillation stability of the cortex to thalamic inputs. The distribution ranges of τ_{pv} (5–14 ms) and τ_{sst} (18–24 ms) are

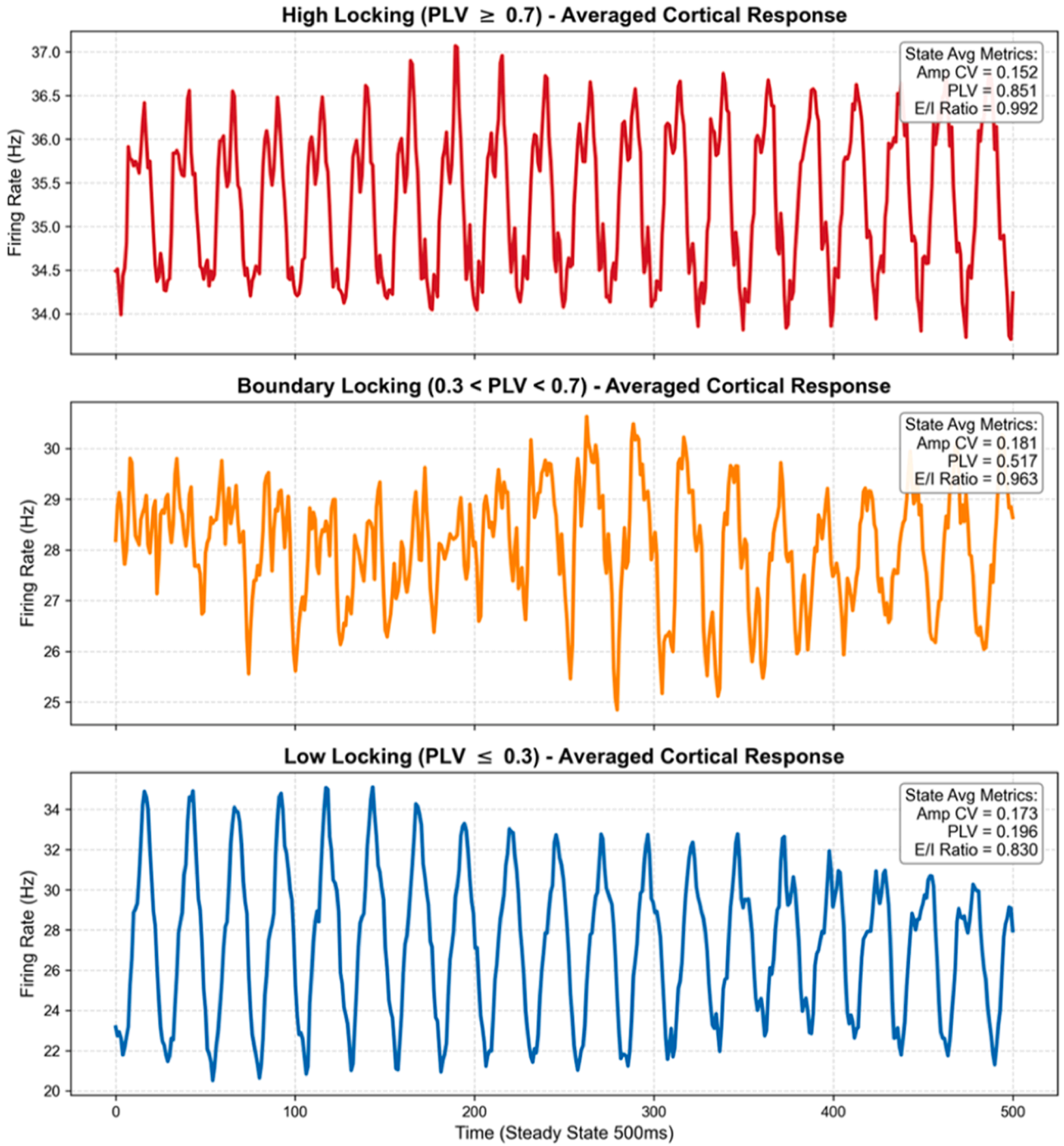


Fig. 3. Cortical ASSR dynamics across thalamocortical 40 Hz PLV.

relatively wide and do not affect core performance, which is consistent with the functional division in physiological experiments that "PV is responsible for fast inhibition and SST is responsible for slow inhibition". Through complementary temporal dynamics, the two jointly regulate the oscillation of Pyr, and can maintain the stability of ASSR without strictly fixing the time constants.

Under the boundary condition of the comprehensive performance of 40 Hz ASSR with $\tau_{pyr} = 15$ ms, Fig. 4 (c) further reveals the regulatory rules of the cortical time constants of PV and SST on thalamocortical phase synchronization by fixing τ_{pyr} . The figure clearly shows that τ_{sst} has a significant threshold effect on the PLV between Pyr and TCR: when $\tau_{sst} > 20$ ms, PLV shows an obvious upward trend, reaching more than 0.98 at the highest, indicating that a sufficiently large τ_{sst}

(corresponding to the slow inhibitory function of SST) is the key to maintaining stable phase synchronization; while when τ_{pv} is distributed within the physiological constraint range, PLV always maintains a high level without significant fluctuations, reflecting the flexible adaptation of the fast inhibitory function of PV to the time constant. It is worth noting that when the cortical time constants combination of $\tau_{pyr} = 15$ ms, $\tau_{pv} = 5$ ms, and $\tau_{sst} = 21$ ms (consistent with Table 2), the model achieves the highest thalamocortical PLV and the smallest CV. This result indicates that the effective parameter space of ACT-NMM is not a random combination, but concentrated in the stable region within the physiological constraint range (consisted with Table 1), and the regulatory role of core parameters is highly consistent with the dynamic rules presented in the figure, which further enhances the biological

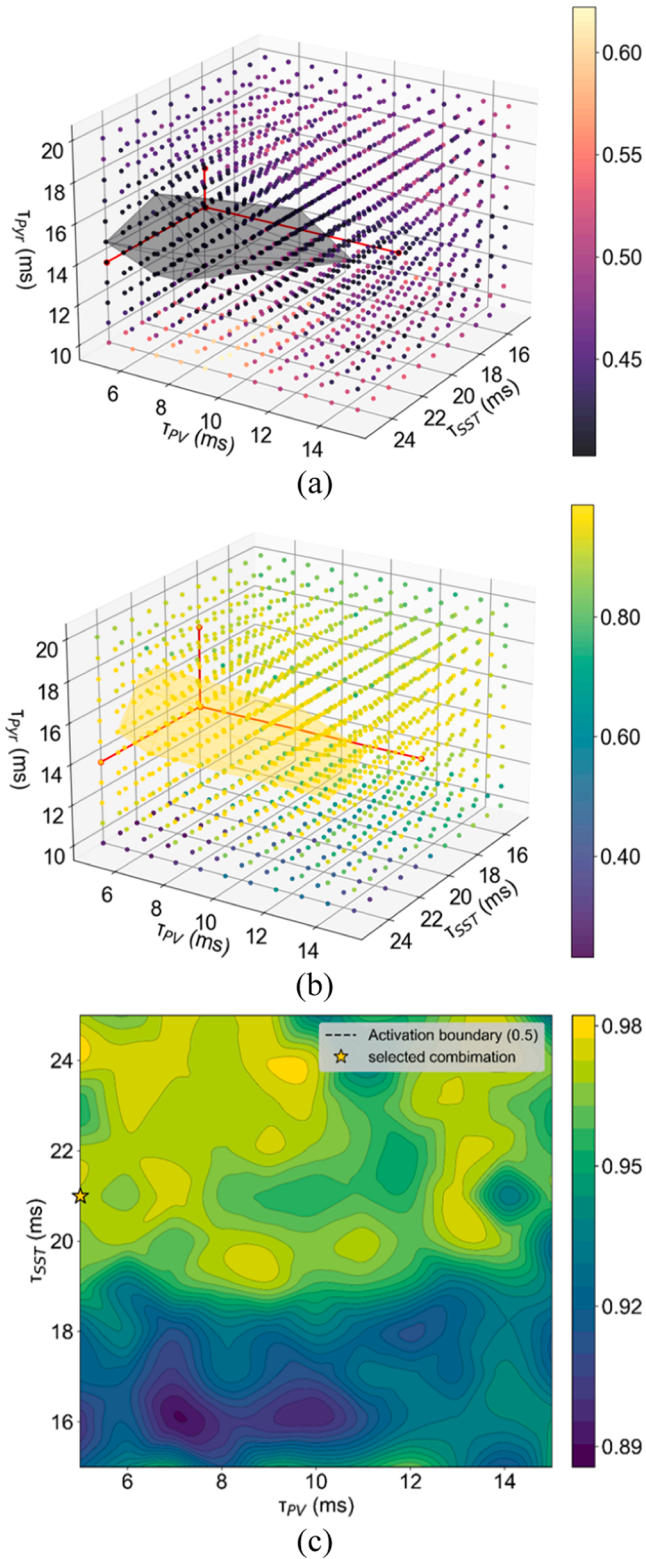


Fig. 4. (a) The CV of fr_{Pyr} amplitude and (b) the thalamocortical PLV for different cortical time constant combinations. (c) The PLV bifurcation of $fr_{TCR} - fr_{Pyr}$ when $\tau_{Pyr} = 15$ ms.

rationality and credibility of the model. It should be noted that the thalamocortical PLV reported here is calculated between internal neural-mass populations (Pyr and TCR) rather than between scalp EEG signals and auditory stimuli. Therefore, its absolute magnitude should be interpreted as a model-based synchronization index reflecting

intrinsic corticothalamic phase alignment, rather than a direct prediction of experimentally measured EEG PLV values.

3.2. Nonlinear dynamical analysis of cortical 40 Hz ASSR under periodic and nonperiodic stimuli

To verify the ability of ACT-NMM to reproduce the phenomenological characteristics of 40 Hz ASSR under the selected parameter combination ($\tau_{Pyr} = 15$ ms, $\tau_{PV} = 5$ ms, $\tau_{SST} = 21$ ms, $w_{thal-ctx} = 3.0$, and $w_{ctx-thal} = 2.0$), we conducted a systematic analysis of its dynamic responses under 40 Hz chirp and OU process noise stimulation based on the ASSR waveform stability calculated by fr_{Pyr} , the frequency-locking ability of target frequency, and the phase-locking ability of $fr_{Pyr} - fr_{TCR}$ and $fr_{Pyr} - Stimulus$. To validate the specificity and reliability of ACT-NMM in reproducing 40 Hz ASSR, we evaluated the phenomenological characteristics of the model under the selected parameter combination by comparing responses to 40 Hz chirp (periodic stimulus) and OU process noise (non-periodic stimulus). As presented in Fig. 5, the numerical simulation results under 40 Hz chirp and OU process noise are significantly different. ACT-NMM exhibits a stable dynamic response to periodic stimulation and has a robust frequency-locking ability to 40 Hz. Consistent with Table 3, Fig. 6 (a) shows that the ASSR waveform under 40 Hz chirp stimulation has significantly lower amplitude variability ($CV=0.397\pm0.05$), maintaining consistent oscillatory amplitude throughout the stimulation period; in contrast, the waveform under OU process noise is highly irregular with larger amplitude fluctuations ($CV=0.782\pm0.11$), failing to form stable periodic oscillations. In Fig. 5 (b), the time-frequency energy map of the 30-50 Hz sub-band reveals under 40 Hz chirp, energy is strongly concentrated at the 40 Hz frequency, showing continuous and stable energy distribution over time; while under OU process noise, the limited energy at 40 Hz dissipates to adjacent frequency bands, with no persistent energy focus. Fig. 6 (c) demonstrates that under 40 Hz chirp stimulation, the PLV of Pyr-Stimulus and Pyr-TCR are all higher than 0.9, indicating robust phase synchronization. Under OU process noise, both PLV of Pyr-Stimulus (0.81 ± 0.19) and Pyr-TCR (0.70 ± 0.25) are obviously decreased, and the phase relationship becomes unstable.

We used r_a to simplify the dynamic of the membrane potential, which reflects the interaction between three cortical populations. As illustrated in Fig. 6 (a), under 40 Hz chirp stimulation, PV rapidly exert inhibitory effects on Pyr firing activity in the early stage of stimulation, which is consistent with their role in fast inhibition. With the continuous accumulation of SST firing, the inhibitory effect of PV on Pyr gradually weakens; in the later stage of stimulation, SST neurons maintain the steady-state firing of Pyr through slow inhibitory regulation, successfully simulating the adaptive firing process of cortical neurons in response to periodic auditory inputs. In contrast, Fig. 6 (b) shows that under OU process noise stimulation, SST firing accumulates rapidly in the early stage, leading to premature and persistent inhibition of Pyr that prevents Pyr from entering the steady-state firing mode required for the generation of 40 Hz ASSR. These dynamic differences in neuronal interactions directly explain why ASSR can be stably reproduced under 40 Hz chirp stimulation. Under OU process noise, the 40 Hz response of the cortex almost entirely stems from the intrinsic dynamics of the cortical network. The randomness of the stimulus signal seriously disrupts the stability and periodicity of the intrinsic firing process in the cortical Pyr-PV-SST neural circuit. Collectively, these results confirm that ACT-NMM specifically exhibits stable dynamic responses to periodic 40 Hz stimulation, with robust frequency-locking ability to the target frequency, while failing to generate meaningful 40 Hz ASSR under non-periodic noise stimulation, which is consistent with neurophysiological observations that ASSR is inherently a phase-locked response induced by periodic auditory inputs.

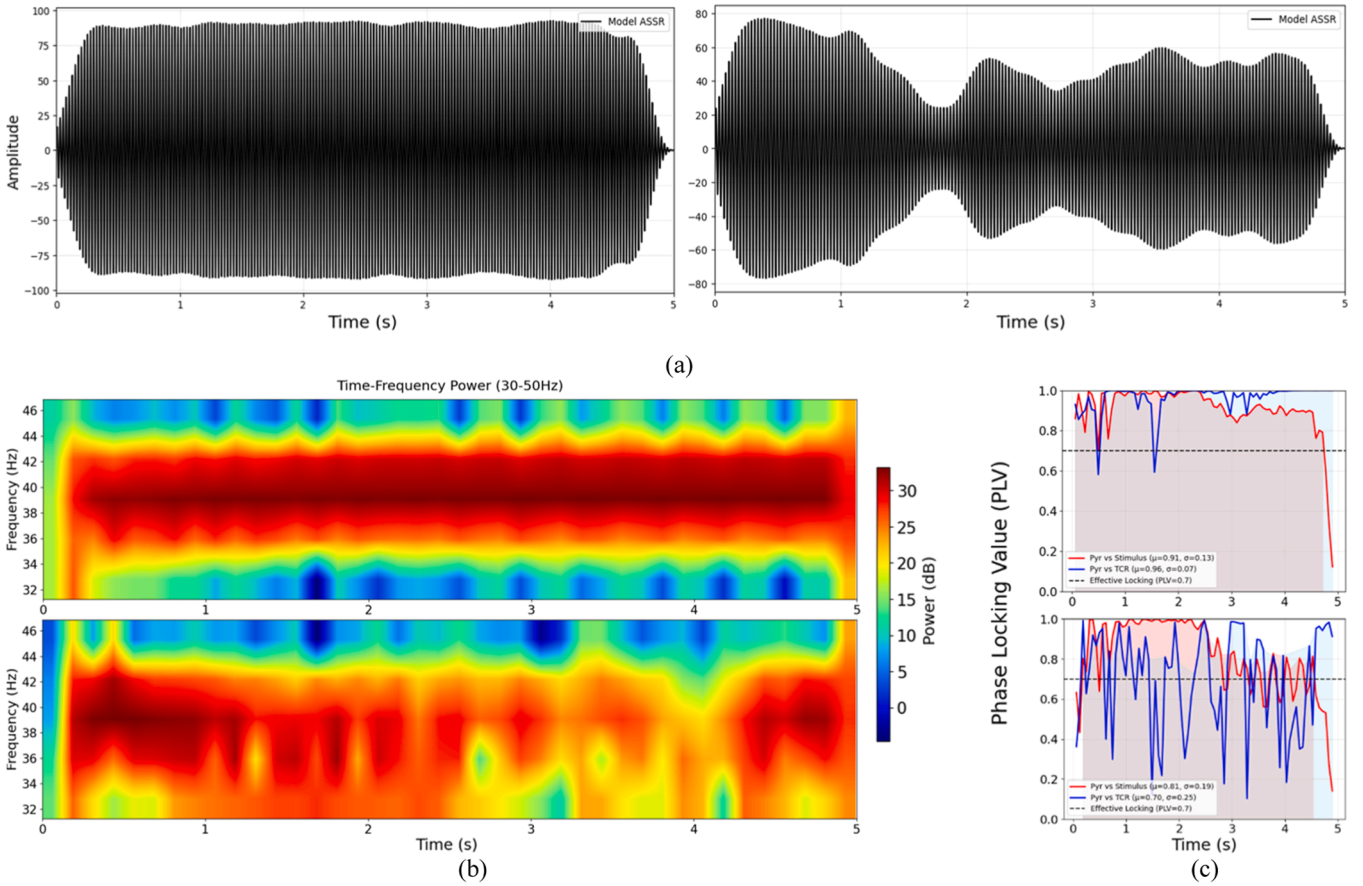


Fig. 5. A comparison of the numerical simulation results for the ACT-NMM under 40 Hz chirp and OU process noise inputs. (a) The ASSR waveform under 40 Hz chirp (top, with lower amplitude variability) and OU process noise (bottom, with greater amplitude variability); (b) The time-frequency energy map of 30–50 Hz under 40 Hz chirp (left, with concentrated energy at the 40 Hz) and OU process noise (right, with energy at 40 Hz dissipating to adjacent frequency bands); (c) The PLV of Pyr-TCR/stimulus under 40 Hz chirp (top, with PLV of Pyr to TCR/input are all higher than 0.9) and OU process noise (bottom, with PLV of Pyr to TCR/input are all lower than 0.9).

Table 3
Comparison of ACT-NMM outputs under different stimulation conditions.

Stimulation	PLV _{Pyr-Stimulus}	PLV _{Pyr-TCR}	CV
40 Hz Chirp	0.91 ± 0.13	0.96 ± 0.07	0.40 ± 0.05
OU process noise	0.81 ± 0.19	0.70 ± 0.25	0.78 ± 0.11

3.3. Phase analysis of the selected parameter combination

The temporal dynamics of selected parameter combination were further characterized by 3D delay embedding analysis. As shown in Fig. 7, this visualization reconstructs f_{Pyr} into a 3-dimensional state space using six different time lags. All time lags consistently exhibit a regular toroidal attractor—a hallmark of periodic dynamical behavior. This morphological feature directly confirms that the coupled thalamocortical system operates in a stable periodic regime under the optimal parameter combination. The toroidal structure remains compact and well-defined across all lags. For short lags ($\tau = 2, 4$ ms), the dense trajectory reflects the rapid inhibitory regulation of PV ($\tau_{pv} = 5$ ms) that stabilizes the short-term dynamics of Pyr. At $\tau = 2$ ms, the point cloud is relatively stretched, forming a "slender cluster" with a certain thickness. When $\tau = 4$ ms, a relatively clear ring structure appears, close to a single-loop attractor (similar to the projection of a limit cycle/quasi-periodic torus disturbed by noise). For intermediate lags ($\tau = 6, 8$ ms), the clearest toroidal contour emerges, matching 1/4–1/3 of the 40Hz oscillation period and verifying the optimal time lags scale of cortico-thalamic coupling. After $\tau = 6$ ms, the ring gradually contracts and thins,

the main axis of the point cloud shrinks to one side, and the thickness decreases. For long lags ($\tau = 10, 12$ ms), it almost collapses into a low-dimensional "flat sheet/line", indicating that the trajectory exhibits high correlation and near-single degree of freedom under longer delays. The maintained regularity of the attractor demonstrates the robustness of the system, attributed to the slow inhibitory modulation of SST ($\tau_{sst} = 21$ ms) that sustains the E-I balance and prevents amplitude drift.

To analyze the modulation effect of two types of thalamic neural clusters on cortical phases, we demonstrated the phase modulation effect of TCR/TRN on Pyr. Fig. 8 (a) shows the phase difference ($\Delta\phi$) between Pyr-TCR and Pyr-TRN rapidly converges to a narrow range within several hundred milliseconds and remains bounded thereafter, indicating a stable phase-locked relationship rather than phase drift or random walk behavior. The unimodal and sharply peaked distribution of $\Delta\phi$ in Fig. 8 (b) further supports strong phase entrainment. In contrast to noise-driven oscillations, which typically yield broad or multimodal phase distributions, the observed concentration reflects robust synchronization. Together with the toroidal attractor observed in the delay-embedded state space, these results indicate that the cortical 40 Hz response corresponds to a stable periodic solution whose phase is consistently regulated by thalamic input.

The time delay embedding portrait and corticothalamic phase difference analysis shows that the periodic dynamic behavior of cortico-thalamic coupling collectively tunes the resonance between the system and the 40Hz stimulation. The regular annular attractors under different time delays not only verifies the biological rationality of the selected parameter combination, but also confirms the stability of the system.

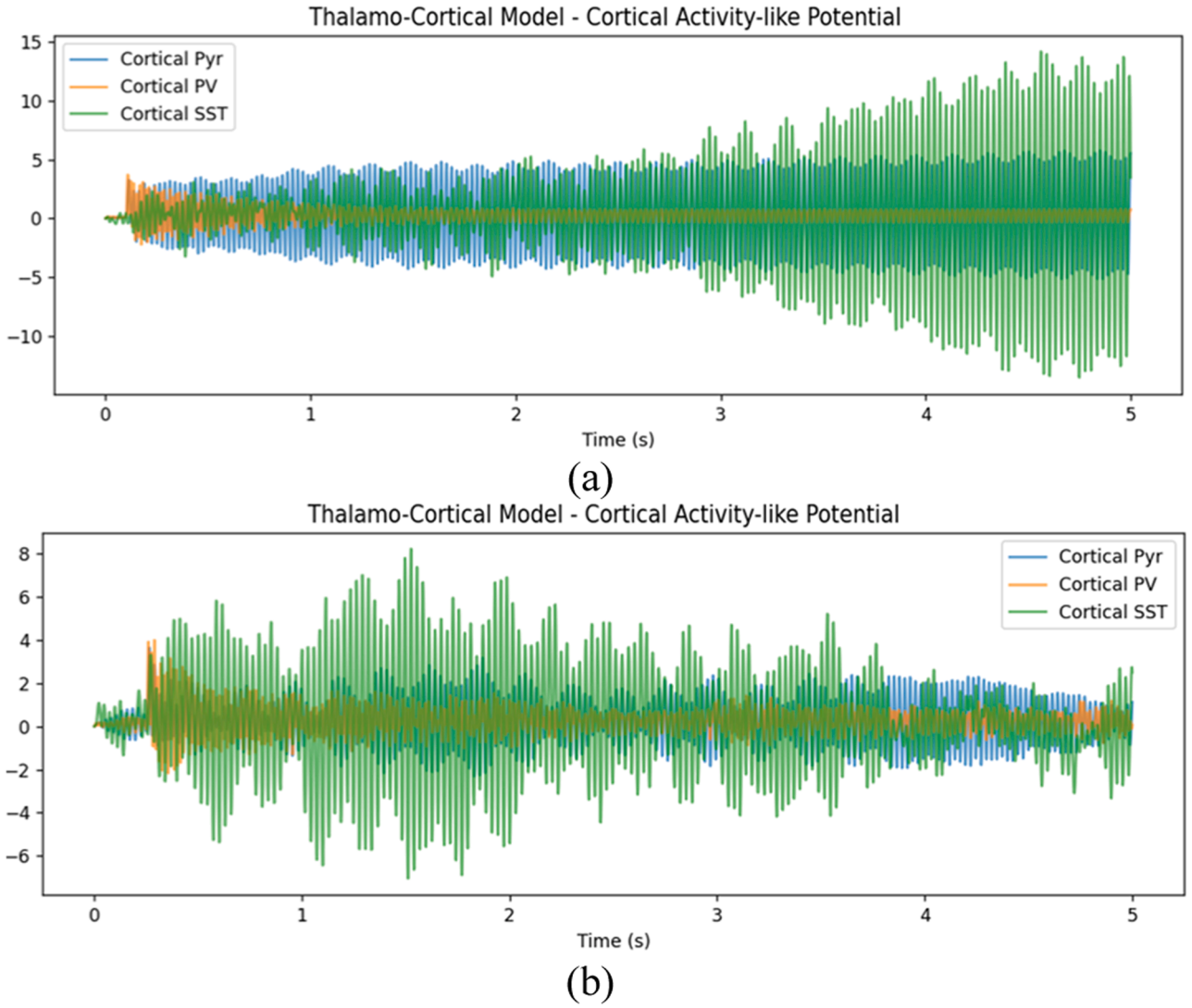


Fig. 6. Dynamic of cortical activity-like potential under 40 Hz chirp and OU process noise. (a) 40 Hz chirp; (b) OU process noise.

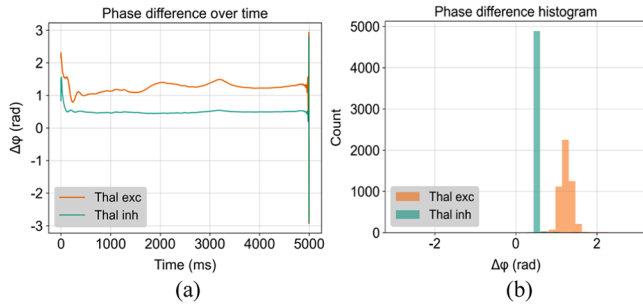


Fig. 8. Corticothalamic phase difference ($\Delta\phi$). (a) The $\Delta\phi$ of Pyr-TCR and Pyr-TRN (b) The phase difference histogram.

The stable phase difference and concentrated phase entrainment in the corticothalamic system provide direct evidence of nonlinear dynamics for the thalamic relay to modulate cortical phases.

4. Discussion

4.1. Response to chirp stimuli of different frequencies

To verify the performance of ACT-NMM to respond to periodic stimuli of different frequencies, comparative analysis was conducted using 20 Hz and 30 Hz chirp. As shown in Fig. 9, the power spectrum analysis revealed the ASSRs effective clusters of cortical time constants for 20 Hz ($\tau_{Pyr} = 36\text{ms}$, $\tau_{PV} = 15\text{ms}$, and $\tau_{SST} = 60\text{ms}$) and 30 Hz ($\tau_{Pyr} = 25\text{ms}$, $\tau_{PV} = 10\text{ms}$, and $\tau_{SST} = 40\text{ms}$). Under multi-frequency chirp stimulation, power spectrum peaks corresponding to the stimulation frequency can appear, thus verifying the ability of frequency-locking to periodic auditory stimulus in numerical simulations. Combined with the PLV curve, ACT-NMM exhibits significantly superior phase-locking and frequency-locking capabilities for 40 Hz compared to 20 Hz and 30 Hz. Consistent phenomena were observed under 20 Hz, 30 Hz, and 40 Hz chirp stimulation: the power spectra of Pyr and TCR showed a main peak corresponding to the stimulation frequency, and simultaneously exhibited secondary peaks at integer multiples of that frequency. The 40 Hz stimulation had the highest main peak power and the most prominent secondary peak at 80 Hz. Although 20 Hz and 30 Hz stimulation displayed the same main peak-secondary peak pattern, both the main peak power and phase-locking ability were significantly weaker than those of 40 Hz, clearly demonstrating the specific response advantage of

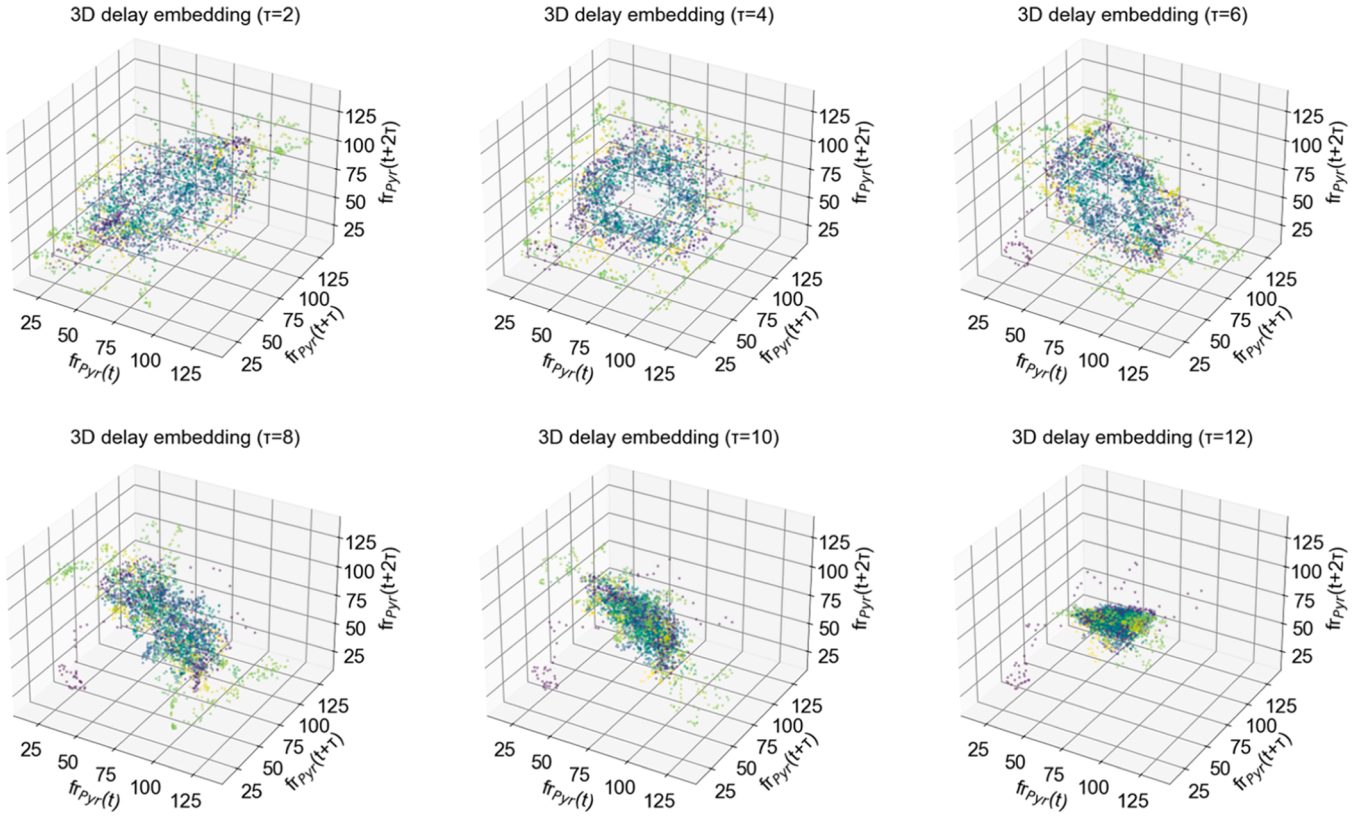


Fig. 7. The fr_{pyr} portrait of 3D delay state space. Where τ is the time lag of delay embedding.

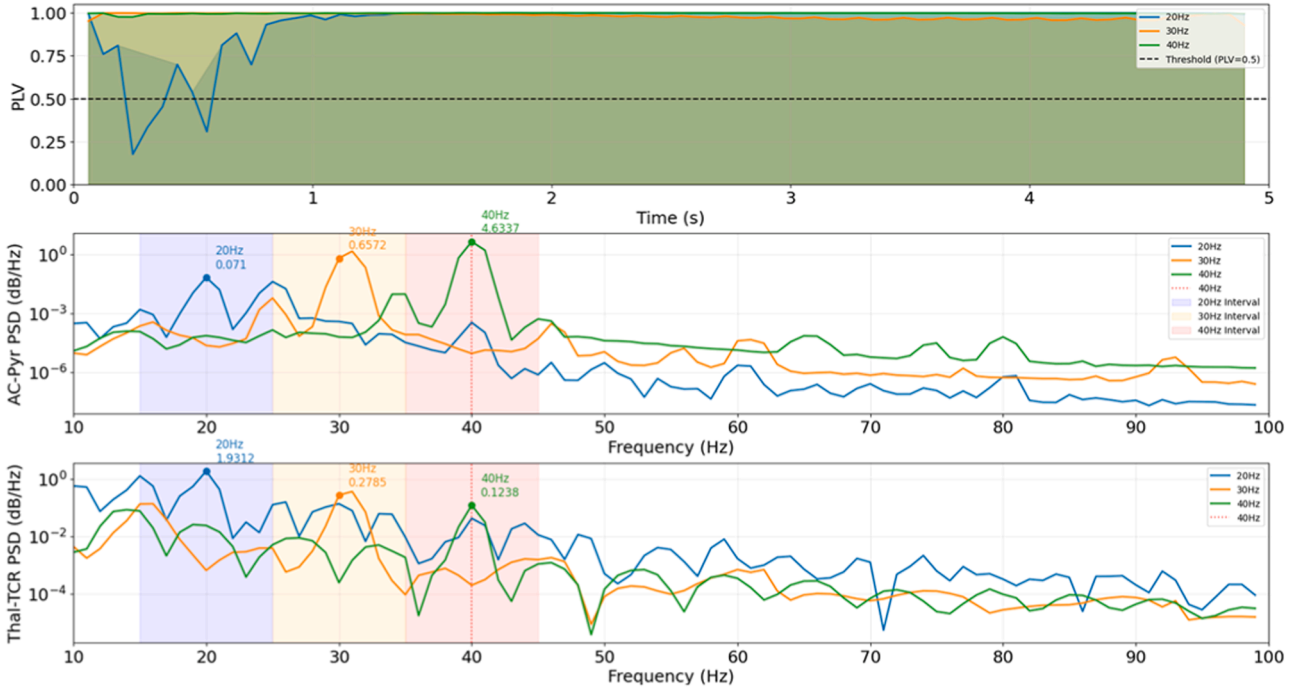


Fig. 9. Corticothalamic procession in response to chirp stimulus of different frequencies. The top figure is the PLV curve of Pyr-Stimulus, the middle figure is the power spectrum curve of Pyr, and the bottom figure is the power spectrum curve of TCR.

ACT-NMM to 40 Hz chirp stimulation.

To further quantify the specificity of 40 Hz ASSR and verify why it exhibits the strongest response among different stimulation frequencies, we analyzed the cortical ASSR performance under different chirp stimuli

(see Table 4). The resonance tuning curve is based on PSD peaks. The 40 Hz stimulus achieves the highest PSD peak, significantly exceeding those at 20 Hz and 30 Hz. This is due to the optimal match between the model's parameter combination and the auditory system's intrinsic resonance,

Table 4

Response to different frequencies stimulation.

Parameter	20 Hz	30 Hz	40 Hz
PSD peak	0.071	0.657	4.634
PLV _{Pyr-TCR}	0.869	0.967	0.983
PLV _{TCR-Stimulus}	0.887	0.948	0.991
CV	0.496	0.481	0.397

which enables the efficient amplification of 40 Hz oscillations via the 'thalamic relay–cortical amplification' mechanism. For the global PLV of Pyr-TCR and TCR-Stimulus, the global phase-locking level gradually increases and peaks at 40 Hz as the stimulation frequency rises. This advantage arises from the thalamus's precise phase preprocessing and suitable cortical time constants. In terms of amplitude stability, the 40 Hz ASSR exhibits the lowest CV, which is prominently lower than the CVs at 20 Hz and 30 Hz. All three metrics consistently confirm that the 40 Hz ASSR has the best response level and outperforms other

frequencies. The results demonstrate that ACT-NMM reproduces a robust 40 Hz ASSR and responds to other frequencies in a physiologically plausible manner. The superior synchronization and frequency locking at 40 Hz confirm the frequency-selective resonance of the corticothalamic circuit.

4.2. ASSR phenomenology analysis of isolated nodes

To analyze the subthreshold dynamics of thalamic nodes and cortical nodes, a bifurcation analysis of isolated nodes was conducted from two aspects: the ASSR activation state and the phase-locking level of excitatory neurons to the stimulus. Dynamic analysis of the ASSR activation state and PLV of the ion conductance ($g_h - g_{LK}$) and time constant ($\tau_{TCR} - \tau_{TRN}$) of the isolated thalamic node found that the thalamic node can hardly generate ASSR. As shown in Fig. 10 (a)(b), the PLV of the thalamus to stimulation changes significantly with the value of ion conductance, and the value of g_{LK} has a more significant impact on PLV

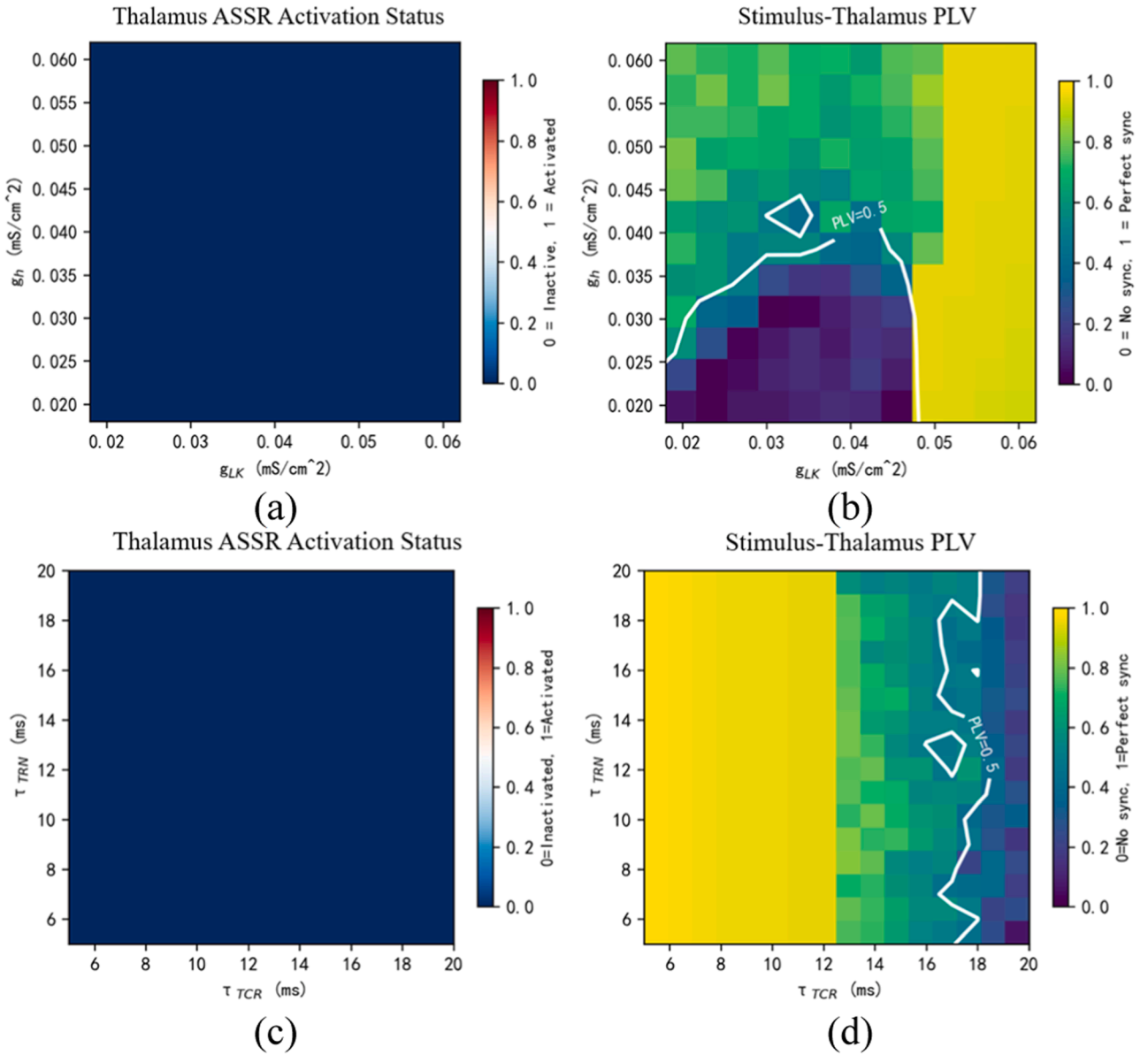


Fig. 10. The bifurcation of thalamic-only model. (a) Ion conductance sensitivity analysis of the ASSR activation state of isolated thalamic node. (b) Ion conductance sensitivity analysis of the PLV of isolated thalamic node. (c) Time constant sensitivity analysis of the ASSR activation state of isolated thalamic node. (d) Time constant sensitivity analysis of the PLV of isolated thalamic node. The white lines and non-rectangular boxes in the graphs represent the boundary of high phase-locking ability with PLV = 0.7.

than g_h . When $g_{LK} < 0.5$, the stimulus-thalamus PLV is less than 0.7 regardless of the value of g_h , which also provides a basis for selecting the ion conductance values ($g_h = 0.40$, $g_{LK} = 0.51$). As shown in Fig. 10 (c) (d), the mathematically feasible solutions ($\tau_{TCR} = 5 - 10$ ms, $\tau_{TRN} = 10 - 20$ ms) with the high PLV conflicts with physiological constraints. It can be seen from the high PLV sub-region that meets biological rationality constraints that the high phase-locking ability of the thalamic node mainly originates from the firing activity of the TCR neuron population. Dynamic analysis shows that the TCR neuron population has a good phase-locking ability to external stimulus signals, but the isolated thalamic node usually cannot generate stable and detectable ASSR signals, with weak response intensity and lack of dynamic stability to continuously maintain high-frequency synchronization, which is more of "preprocessing" and "synchronization before transmission" of input signals.

The ASSR activation state and phase-locking ability of the isolated auditory cortical node in the parameter space of the time constant of the inhibitory populations ($\tau_{PV} - \tau_{SST}$) were analyzed. As shown in Fig. 11, the cortical node can generate ASSR under almost all combinations of time constant after selecting the boundary condition of $\tau_{Pyr} = 15$ ms. To find the ASSR activation boundary, the value range of $\tau_{PV} = 1 - 15$ ms and that of $\tau_{SST} = 11 - 25$ ms. The firing activity of Pyr in the cortex after receiving external stimulation is significantly stronger than that of thalamic TCR, but its phase-locking ability to input stimulation is poor. However, in the evaluation of the overall PLV level in the parameter space, the isolated cortical node (0.52 ± 0.24) is significantly lower than the isolated thalamic node (0.74 ± 0.25), indicating that the phase-locking ability of ACT-NMM to stimulation mainly originates from the neural activity of the thalamic node. Dynamic analysis shows that the auditory cortex is the main generation site of ASSR, and the thalamus provides key input for it through ascending projections. The neuron populations of the cortex node can further amplify and stabilize the periodic signals transmitted by the thalamus, and form continuous high-frequency synchronous oscillations (such as 40 Hz ASSR) through the intrinsic dynamic characteristics of the cortical neuron network. Although the isolated thalamic node can perform synchronized preprocessing of stimuli through the TCR-TRN circuit, it cannot generate detectable ASSR and has a high frequency deviation (1.75 ± 2.10 Hz) due to the lack of stability to continuously maintain frequency

synchronization. However, the isolated cortical node has good frequency-locking ability (frequency deviation: 0.39 ± 0.37 Hz) and extensively activates ASSR, despite having weak phase-locking ability.

The ACT-NMM combines the advantages of thalamic nodes and cortical nodes. Through the precise phase-locking and relay transmission functions provided by the thalamus, the cortical nodes with good frequency-locking ability receive a purer 40 Hz periodic oscillation. The persistence of periodic cortical oscillations under low-locking conditions indicates that the cortical Pyr-PV-SST microcircuit is capable of sustaining intrinsic oscillatory activity. In contrast, increasing thalamocortical locking primarily improves phase stability and synchronization rather than creating oscillatory activity *de novo*. These findings suggest that the cortex acts as the principal generator of the 40 Hz ASSR, whereas the thalamus mainly serves as a synchronization and relay modulation system. This interpretation is consistent with the dynamical transition observed from low-locking to high-locking states, in which thalamocortical coupling progressively stabilizes and synchronizes cortical oscillations without fundamentally changing their oscillatory nature.

The present results are also consistent with recent neural dynamics studies emphasizing that stable behavior of brain-inspired models depends on the interaction between local E-I regulation and cross-region coupling. In particular, the E-I homeostasis mechanism highlighted by Wang et al. (Wang et al., 2025) supports our interpretation that PV/SST-mediated inhibitory regulation is essential for maintaining stable cortical oscillations. The cortical-subcortical modeling framework of Cui et al. (Cui et al., 2025) and the visual-cerebellar modeling strategy of Li et al. (Li et al., 2026) further illustrate the value of decomposing complex neural functions into physiologically meaningful interacting modules, which is the same rationale underlying the present TCR-TRN and Pyr-PV-SST coupling design.

4.3. Sensitivity to corticothalamic projection delay

Because ACT-NMM is formulated as a delay-coupled corticothalamic system, the long-range projection delay may influence the phase relationship between cortical and thalamic populations. To evaluate whether the 13 ms delay used in the main simulations critically determines the model behavior, we performed a sensitivity analysis by

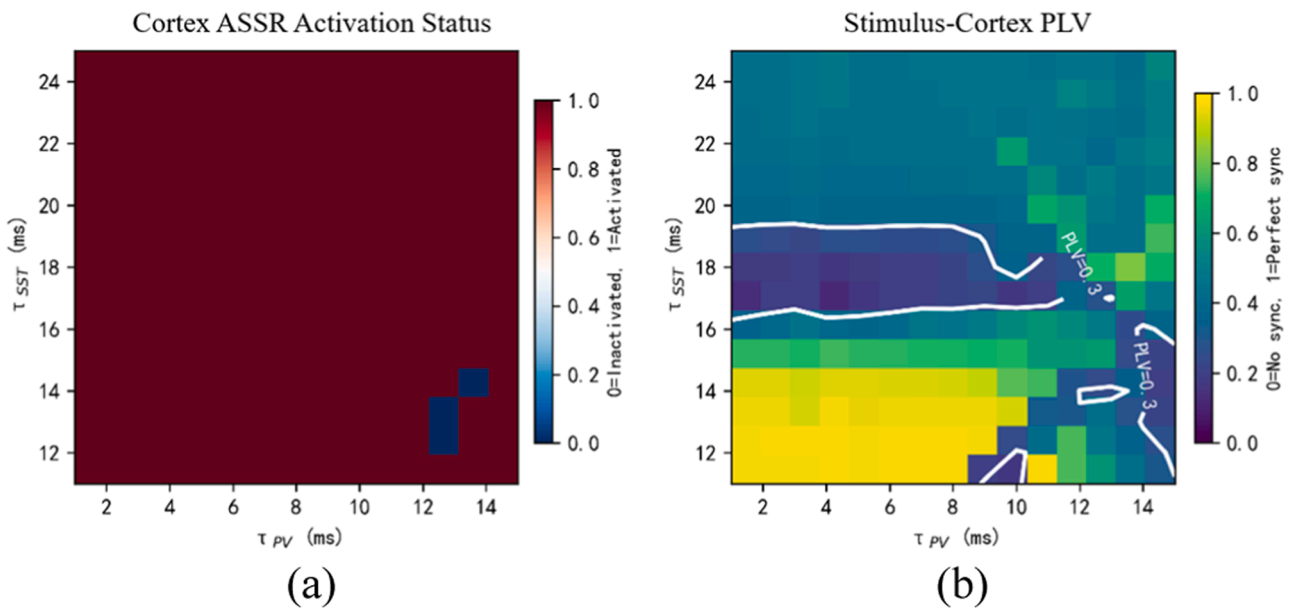


Fig. 11. The bifurcation of cortical-only model. (a) Time constant sensitivity analysis of the ASSR activation state of isolated cortical node. (b) Time constant sensitivity analysis of the PLV of isolated cortical node. The white lines and non-rectangular boxes in the graphs represent the boundary of high phase-locking ability with $PLV = 0.7$.

symmetrically varying the corticothalamic and thalamocortical delays from 5 to 25 ms.

As shown in Fig. 12, all of the tested delay conditions resulted in detectable 40 Hz ASSR activation; however, the quality of thalamocortical synchronization varied substantially with delay. Around the reference value, the model remained in a stable high-locking regime. Notably, the 13 ms condition produced high Pyr-TCR synchronization and zero frequency deviation from 40 Hz. Neighboring delays from 11 to 16 ms also maintained Pyr-TCR PLV above 0.7, indicating that moderate delay fluctuations do not destroy the periodic ASSR regime. In contrast, shorter delays ranging from 5 to 10 ms markedly reduced Pyr-TCR phase locking, although the 40 Hz target power ratio remained relatively high. This suggests that the projection delay not only determines whether a 40 Hz spectral component can be generated, but also controls whether thalamic relay activity arrives at the cortex with an effective phase relationship for stabilizing cortical oscillations. Therefore, the 13 ms delay adopted in the main analysis should be interpreted as a physiologically plausible reference value located within a robust synchronization range, rather than as a narrowly tuned parameter. These findings further support the interpretation that stable 40 Hz ASSR generation in ACT-NMM depends on coordination among cortical intrinsic oscillatory dynamics, thalamic relay modulation, and delayed recurrent corticothalamic feedback.

5. Conclusion

We propose the ACT-NMM, an auditory corticothalamic neural mass model incorporating a closed-loop interaction between the cortical Pyr-PV-SST and the thalamic TCR-TRN circuit. A parameter sensitivity

analysis was performed to identify the selected parameter combination simultaneously satisfies the canonical ASSR phenomenological characteristics and existing physiological constraints. Through the numerical simulation, we quantitatively demonstrate the thalamocortical process underlying ASSR generation at the NMMs level for the first time. In summary, the neural dynamics of ACT-NMM shows that cortical 40 Hz ASSR are organized in a manner that is both dynamically stable and informationally structured. This integrated perspective provides a principled computational framework for organizing ASSR-related modeling results and may serve as a reference for future studies aiming to link observed neural dynamics with underlying circuit parameters.

Several limitations should also be acknowledged. First, ACT-NMM simplifies some ion-current mechanisms and does not aim to reproduce microscopic neuronal phenomena in full biophysical detail. Second, the pyramidal firing rate used in this study is an internal neural-mass variable and should not be interpreted as a direct prediction of experimentally measured LFP or EEG amplitude. Although measurable electrophysiological signals such as LFP, EEG, and MEG are closely related to synaptic currents in geometrically organized pyramidal neurons (Cook et al., 2022) (Einevoll Kayser and Logothetis, 2013) (Deco et al., 2008), the present study does not explicitly model the forward transformation from neural-mass activity to these signals. Third, although ACT-NMM includes nonlinear intrinsic population dynamics and a state-dependent thalamic input transmission gate, the long-range bidirectional corticothalamic projections are still represented by additive coupling terms. This simplified formulation improves interpretability and enables systematic parameter sensitivity analysis, but it cannot fully capture the nonlinear gain-control mechanisms observed in real thalamocortical circuits. Future work will therefore focus on incorporating neural-mass-to-LFP/EEG forward modeling, observation noise, empirical calibration with electrophysiological recordings, and nonlinear state-dependent corticothalamic coupling functions. These extensions may help narrow the gap between theoretical simulations and measured ASSR signals while preserving the mechanistic interpretability of the current framework. Furthermore, recent advances in data-driven representation learning and multimodal analysis have the potential to provide valuable tools for EEG/LFP signal quality assessment, multimodal electrophysiological integration, and calibration between model-derived neural dynamics and empirical recordings (Shen Zhou and Chen, 2025) (Lan Xian and Zhou, 2025) (Zheng Zhou and Shang, 2025) (Zhou Li and Yang, 2025) (Zhou Zhao and Luo, 2023) (Song Zhou and Luo, 2024) (Zhou Lan and Wei, 2022) (Zhou Shen and Wei, 2025) (Cheng Song and Zhou, 2024) (Lang Zhou and Wei, 2025) (Zhou Li and Wei, 2025).

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CRediT authorship contribution statement

Mu Zhu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Xiaoya Liu:** Writing – review & editing, Visualization, Validation, Supervision. **Xinmeng Guo:** Writing – review & editing, Validation, Supervision, Formal analysis. **Yilin Zhang:** Investigation. **Ziliang Cai:** Writing – review & editing, Visualization, Investigation. **Shuang Liu:** Writing – review & editing, Validation, Supervision. **Dong Ming:** Writing – review & editing, Supervision.

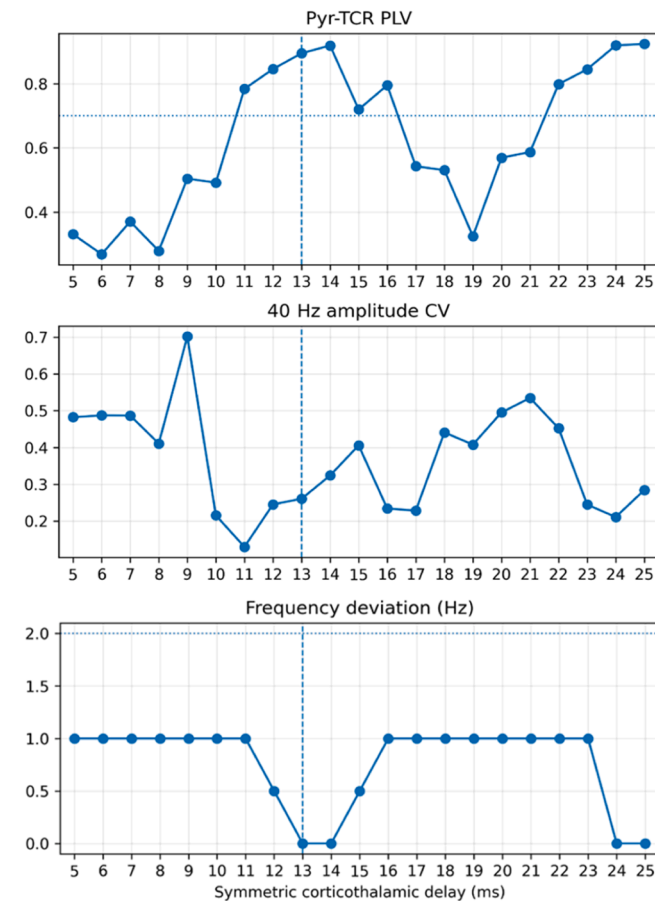


Fig. 12. Delay sensitivity analysis of ACT-NMM. The three graphs from top to bottom show the dynamic characteristic curves of Pyr-TCR PLV, 40 Hz amplitude CV and frequency deviation under a delay ranging from 5 to 25 ms.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Alexander, G M, Fisher, T L, & Godwin, D W (2006). Differential response dynamics of corticothalamic glutamatergic synapses in the lateral geniculate nucleus and thalamic reticular nucleus. *Neuroscience*, 137(2), 367–372.
- Aponte, D. A., Handy, G., Kline, A. M., et al. (2021). Recurrent network dynamics shape direction selectivity in primary auditory cortex. *Nature Communications*, 12, 314.
- Beiran, Manuel, & Litwin-Kumar, Ashok (2025). Prediction of neural activity in connectome-constrained recurrent networks. *Nature Neuroscience*, 28, 2561–2574.
- Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature Neuroscience*, 20(3), 340–352.
- Cauler, L. J., & Connors, BW. (1994). Synaptic physiology of horizontal afferents to layer I in slices of rat SI neocortex. *The Journal of Neuroscience : the Official Journal of the Society for Neuroscience*, 14, 751–762.
- Cheng, S., Song, J., Zhou, M., et al. (2024). Ef-detr: A lightweight transformer-based object detector with an encoder-free neck. *IEEE Trans Ind Inf*, 20(11), 12994–13002.
- Cook, B., Peterson, A., Woldman, W., & Terry, J. (2022). Neural field models: a mathematical overview and unifying framework. *Math Neurosci Appl*, 2, 1–67.
- Cui, L., Yu, Y., Yin, L., Hou, S., & Wang, Q. (2025). Cortical-subcortical neural networks for motor learning and storing sequence memory. *Neural Networks*, 189, 1–16.
- D, Du X, & Jansen, B H (2011). A neural network model of normal and abnormal auditory information processing. *Neural Networks*, 24(6), 568–574.
- Deco, G., Jirsa, V. K., Robinson, P. A., Breakspear, M., & Friston, K. (2008). The dynamic brain: from spiking neurons to neural masses and cortical fields. *PLoS Computational Biology*, 4(8), 1–35.
- Einevoll, G., Kayser, C., Logothetis, N., et al. (2013). Modelling and analysis of local field potentials for studying the function of cortical circuits. *Nature reviews. Neuroscience*, 14, 770–785.
- Gabernet, L., Jadhav, S P, Feldman, D E, et al. (2005). Somatosensory integration controlled by dynamic thalamocortical feed-forward inhibition. *Neuron*, 48(2), 315–327.
- Griskova, I., Morup, M., Parnas, J., et al. (2007). The amplitude and phase precision of 40 Hz auditory steady-state response depend on the level of arousal. *Experimental Brain Research*, 183(1), 133–138.
- Hackett, T. A., Barkat, T. R., O'Brien, B. M. J., Hensch, T. K., & Polley, D. B. (2011). Linking topography to tonotopy in the mouse auditory thalamocortical circuit. *The Journal of Neuroscience : The Official Journal of the Society for Neuroscience*, 31, 2983–2995.
- Halabisky, Brian, et al. (2006). Electrophysiological classification of somatostatin-positive interneurons in mouse sensorimotor cortex. *Journal of Neurophysiology*, 96(2), 834–845.
- Ibrahim, B A, Murphy, C A, Yudinsev, G., et al. (2021). Corticothalamic gating of population auditory thalamocortical transmission in mouse. *eLife*, 10, 1–23.
- Jones, E G (2001). The thalamic matrix and thalamocortical synchrony. *Trends in Neurosciences*, 24(10), 595–601.
- Jones, E. G. (2007). *The thalamus* (2nd ed.) Cambridge, UK: Cambridge University Press.
- Kim, D., Jeong, H., Lee, J., et al. (2016). Distinct roles of parvalbumin- and somatostatin-expressing interneurons in working memory. *Neuron*, 92(4), 902–915.
- King, A. J., & Nelken, I. (2009). Unraveling the principles of auditory cortical processing: can we learn from the visual system? *Nature Neuroscience*, 12, 698–701.
- Koch, C., Rapp, M., & Segev, I. (1996). A brief history of time (constants). *Cerebral Cortex (New York, N.Y. : 1991)*, 6(2), 93–101.
- Lan, X., Xian, W., Zhou, M., et al. (2025). No-reference image quality assessment: exploring intrinsic distortion characteristics via generative noise estimation with mamba. *IEEE Transactions on Circuits and Systems for Video Technology : A Publication of the Circuits and Systems Society*, 35(12), 12692–12706.
- Lang, S., Zhou, M., Wei, X., et al. (2025). Image quality assessment: Exploring the similarity of deep features via covariance-constrained spectra. *IEEE Trans Broadcast*, 72(1), 303–316.
- Li, L., Li, Y., Zhou, M., Tao, H. W., & Zhang, L. I. (2013). Intracortical multiplication of thalamocortical signals in mouse auditory cortex. *Nature Neuroscience*, 16, 1179–1181.
- Li, L.-Y., et al. (2014). Differential receptive field properties of parvalbumin and somatostatin inhibitory neurons in mouse auditory cortex. *Cerebral Cortex*, 25, 1782–1791.
- Li, Y., Lopez-Huerta, V. G., Adiconis, X., Levandowski, K., Choi, S., Simmons, S. K., Arias-Garcia, M. A., Guo, B., Yao, A. Y., Blosser, T. R., et al. (2020). Distinct subnetworks of the thalamic reticular nucleus. *Nature*, 583, 819–824.
- Li, Z., Yin, L., Cui, L., Yu, Y., & Wang, Q. (2026). A biologically inspired neural network for optic flow-based reactive navigation with dual depth encoding. *Bioinspiration & Biomimetics*, 21(2), 1–26.
- Li, J., Li, Z., Wang, X., et al. (2024). The thalamocortical mechanism underlying the generation and regulation of the auditory steady-state responses in awake mice. *The Journal of Neuroscience : The Official Journal of the Society for Neuroscience*, 44(1), 1–14.
- Lien, A. D., & Scanziani, M. (2013). Tuned thalamic excitation is amplified by visual cortical circuits. *Nature Neuroscience*, 16, 1315–1323.
- Liu, C., Zhou, C., Wang, J., Fietkiewicz, C., & Loparo, K. A. (2021). Delayed feedback-based suppression of pathological oscillations in a neural mass model. *IEEE Transactions On Cybernetics*, 51(10), 5046–5056.
- Malik, R., Pai, E L L, Rubin, A N, et al. (2019). Tsc1 represses parvalbumin expression and fast-spiking properties in somatostatin lineage cortical interneurons. *Nature Communications*, 10(1), 1–16.
- Mill, R., Coath, M., Wennekers, T., et al. (2011). A neurocomputational model of stimulus-specific adaptation to oddball and Markov sequences. *PLoS Computational Biology*, 7(8), 1–15.
- Mill, R., Coath, M., Wennekers, T., & Denham, SL. (2011). A neurocomputational model of stimulus-specific adaptation to oddball and Markov sequences. *PLoS Computational Biology*, 7, 1–15.
- Moran, R J, Kiebel, S J, Stephan, K E, et al. (2007). A neural mass model of spectral responses in electrophysiology. *NeuroImage*, 37(3), 706–720.
- Park, Y., & Geffen, MN. (2020). A circuit model of auditory cortex. *PLoS Computational Biology*, 16(7), 1–29.
- Peng, H., Xie, P., Liu, L., Kuang, X., Wang, Y., Qu, L., Gong, H., Jiang, S., Li, A., Ruan, Z., et al. (2021). Morphological diversity of single neurons in molecularly defined cell types. *Nature*, 598, 174–181.
- Pfeffer, C. K., Xue, M., He, M., Huang, Z. J., & Scanziani, M. (2013). Inhibition of inhibition in visual cortex: The logic of connections between molecularly distinct interneurons. *Nature Neuroscience*, 16, 1–15.
- Phillips, Elizabeth AK, Schreiner, Christoph E., & Hasenstaub, Andrea R. (2017). Cortical interneurons differentially regulate the effects of acoustic context. *Cell Reports*, 20(4), 771–778.
- Picton, T. W., John, M. S., Dimitrijevic, A., et al. (2003). Human auditory steady-state responses. *International Journal of Audiology*, 42(4), 177–219.
- Povysheva, N V, Zaitsev, A V, Rotaru, D C, et al. (2008). Parvalbumin-positive basket interneurons in monkey and rat prefrontal cortex. *Journal of Neurophysiology*, 100(4), 2348–2360.
- Reyes, S. A., Lockwood, A. H., Salvi, R. J., et al. (2005). Mapping the 40 Hz auditory steady-state response using current density reconstructions. *Hearing Research*, 204(1–2), 1–15.
- Roux, F., Wibral, M., Singer, W., Aru, J., & Uhlhaas, P. J. (2013). The phase of thalamic alpha activity modulates cortical gamma-band activity: evidence from resting-state MEG recordings. *The Journal of Neuroscience : The Official Journal of The Society for Neuroscience*, 33, 17827–17835.
- Roy, D. S., Zhang, Y., Halassa, M. M., & Feng, G. (2022). Thalamic subnetworks as units of function. *Nature Neuroscience*, 25, 140–153.
- Rubio, Garrido, P., Manzo, Pe rez-de, Porrero, F., Galazo, C., J. M., & Clasca, F. (2009). Thalamic input to distal apical dendrites in neocortical layer 1 is massive and highly convergent. *Cerebral Cortex*, 19, 2380–2395.
- Schellenberger Costa, M., Weigenand, A., Ngo, H. V. V., et al. (2016). A thalamocortical neural mass model of the EEG during NREM sleep and its response to auditory stimulation. *PLoS Computational Biology*, 12(9), 1–20.
- Shen, W., Zhou, M., Chen, Y., et al. (2025). Image quality assessment: Investigating causal perceptual effects with abductive counterfactual inference. In *Proc. IEEE Conf. Comput. Vis. Pattern Recog* (pp. 17990–17999).
- Song, J., Zhou, M., Luo, J., et al. (2024). Boundary-aware feature fusion with dual-stream attention for remote sensing small object detection. *IEEE Transactions On Geoscience and Remote Sensing : A Publication of the IEEE Geoscience and Remote Sensing Society*, 63, 1–13.
- Wang, H., Jia, W., Zhou, Y., Li, Z., Yu, Y., & Wang, Q. (2025). Unveiling the role of excitation-inhibition homeostasis in stroke recovery from individualized whole-brain dynamical modeling. *Communications in Nonlinear Science & Numerical Simulation*, 151, 1–17.
- Wilson, H R, & Cowan, J D (1973). A mathematical theory of the functional dynamics of cortical and thalamic nervous tissue. *Kybernetik*, 13(2), 55–80.
- Zhao, Z., Tang, X., Chen, Y., et al. (2025). A parallel tonotopically arranged thalamocortical circuit for sound processing. *Neuron*, 113(12), 1998–2013.
- Zheng, Z., Zhou, M., Shang, Z., et al. (2025). GAANet: Graph aggregation alignment feature fusion for multispectral object detection. *IEEE Trans Ind Inf*, 21(11), 8282–8292.
- Zhou, M., Lan, X., Wei, X., et al. (2022). An end-to-end blind image quality assessment method using a recurrent network and self-attention. *IEEE Trans Broadcast*, 69(2), 369–377.
- Zhou, M., Li, J., Wei, X., et al. (2025). AFES: Attention-based feature excitation and sorting for action recognition. *IEEE Trans Consum Electron*, 71(2), 5752–5760.
- Zhou, M., Li, Y., Yang, G., et al. (2025). COFNet: contrastive object-aware fusion using box-level masks for multispectral object detection. *IEEE Transactions On Multimedia*, 27, 7444–7458.
- Zhou, M., Shen, W., Wei, X., et al. (2025). Blind image quality assessment: Exploring content fidelity perceptibility via quality adversarial learning. *International Journal of Computer Vision*, 133(6), 3242–3258.
- Zhou, M., Zhao, X., Luo, F., et al. (2023). Robust rgb-t tracking via adaptive modality weight correlation filters and cross-modality learning. *ACM Trans Multimedia Comput Commun Appl*, 20(4), 1–20.